Supplementary Materials

Historical Religious Institutionalization and Population Health:
A Cross-National Panel Analysis

S1 Global Distribution of Key Variables

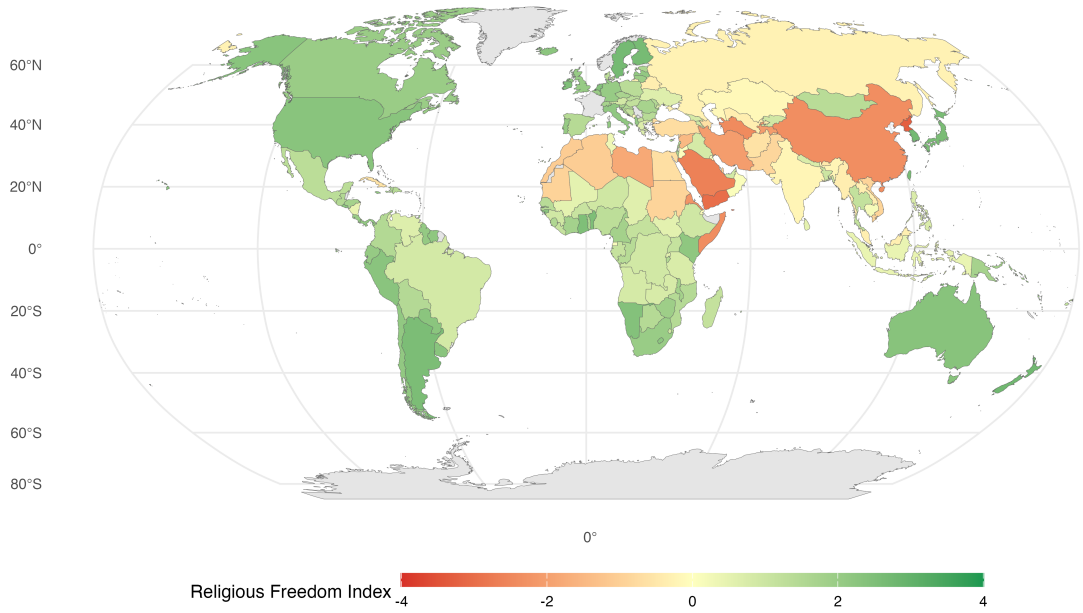

Figure S1: Global distribution of the V-Dem Religious Freedom Index (most recent available year per country, typically 2019–2020). The index ranges from approximately −4 (severe restriction) to +4 (full freedom). Red = restricted; green = free; grey = missing data. Source: V-Dem v15.

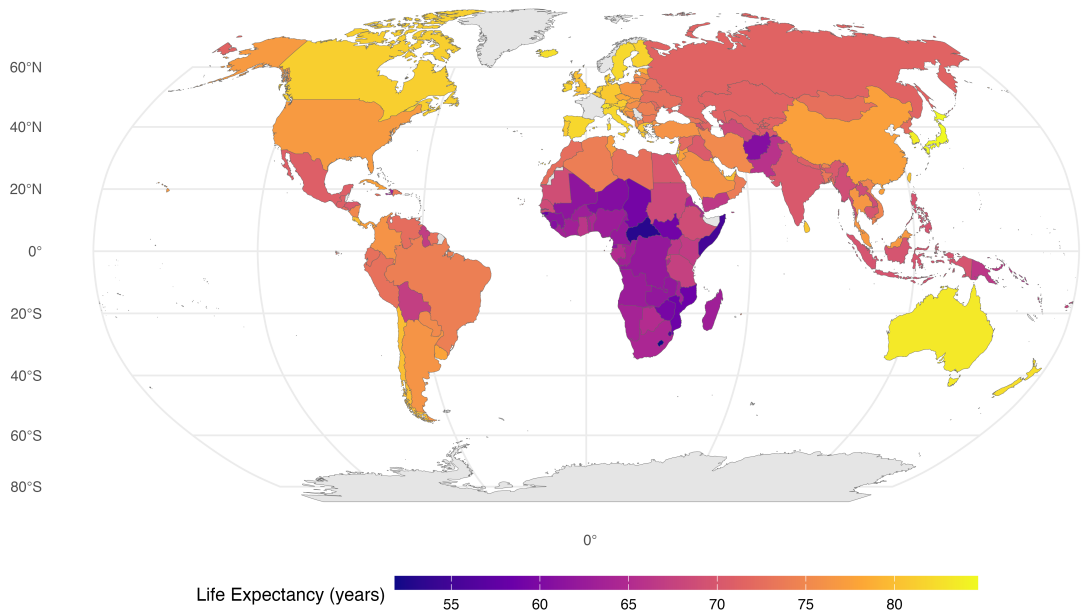

Figure S2: Global distribution of life expectancy at birth (most recent available year per country, typically 2019–2020). Source: WHO Global Health Observatory; QoG Standard Dataset.

S2 Correlation Matrix

Table S1 presents the pairwise Pearson correlation matrix for all variables in the estimation sample. Religious freedom is strongly positively correlated with electoral democracy ($r = 0.72$), freedom of expression ($r = 0.78$), and log GDP per capita ($r = 0.45$). These correlations motivate the mediation analysis in the main text, since they suggest that religious freedom is embedded in a broader cluster of political and economic liberalization. Life expectancy at birth is positively correlated with all institutional and economic variables, with the strongest association observed for log GDP per capita ($r = 0.62$) and electoral democracy ($r = 0.47$).

Table S1: Correlation Matrix: Key Variables in the Estimation Sample

	(1)	(2)	(3)	(4)	(5)	(6)
(1) Life expectancy at birth	1.00					
(2) Religious freedom index	0.41	1.00				
(3) Electoral democracy	0.47	0.72	1.00			
(4) Freedom of expression	0.44	0.78	0.90	1.00		
(5) Log GDP per capita	0.62	0.45	0.47	0.44	1.00	
(6) Urbanization (%)	0.57	0.29	0.28	0.26	0.73	1.00

Notes: Pairwise Pearson correlations across all available country-year observations, 1990–2020. N ranges from 5,201 to 5,327 depending on variable availability. Sources: V-Dem v15; WHO GHO; QoG Standard Dataset.

S3 Causal Mediation Analysis

Table S2 presents the full causal mediation results for six candidate pathways, following Imai et al. (2010). The table reports the average causal mediation effect (ACME), average direct effect (ADE), total effect, and proportion mediated for each candidate mediator. Quasi-Bayesian 95% confidence intervals are based on 1,000 simulations. Country and year fixed effects are absorbed via within-transformation prior to estimation.

Table S2: Mechanism Analysis: Causal Mediation via Two-Way Fixed Effects

	Electoral democracy	Liberal democracy	Freedom of expression	Urbanization (%)	Log GDP per capita	Mean yrs schooling
Panel A: ACME — average causal mediation effect (indirect)						
Estimate	+0.216***	+0.129***	+0.286***	+0.053***	-0.059***	+0.014
	[0.128, 0.304]	[0.035, 0.220]	[0.186, 0.383]	[0.028, 0.079]	[-0.088, -0.032]	[-0.007, 0.037]
Panel B: ADE — average direct effect						
Estimate	+0.337***	+0.443***	+0.270**	+0.603***	+0.624***	+1.043***
	[0.151, 0.535]	[0.246, 0.639]	[0.068, 0.474]	[0.406, 0.792]	[0.442, 0.804]	[0.726, 1.352]
Panel C: Total effect and proportion mediated						
Total effect	+0.552***	+0.572***	+0.556***	+0.655***	+0.565***	+1.056***
	[0.378, 0.742]	[0.384, 0.751]	[0.385, 0.740]	[0.462, 0.848]	[0.387, 0.747]	[0.730, 1.359]
Prop. mediated (%)	39.5	22.2	51.7	8.0	-10.4	1.2
	[22.1, 64.4]	[6.5, 45.1]	[30.8, 83.2]	[4.3, 13.6]	[-17.6, -5.4]	[-0.7, 3.7]
Observations	5,326	5,305	5,327	5,201	5,320	2,720
Simulations	1,000	1,000	1,000	1,000	1,000	1,000

Notes: Causal mediation analysis following Imai et al. (2010). Country and year fixed effects absorbed via within-transformation prior to estimation, 1990–2020. Mediator model: mediator $\sim$ Religious Freedom Index. Outcome model: life expectancy $\sim$ Religious Freedom Index + mediator. ACME = average causal mediation effect; ADE = average direct effect. Quasi-Bayesian 95% confidence intervals in brackets (1,000 simulations). Treatment increment: one unit increase in the Religious Freedom Index. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

S4 Heterogeneity Analysis

Table S3 reports the religious freedom coefficient from separate TWFE regressions within subgroups defined by World Bank income classification, world region, and historical religious majority. Stratified estimation permits the coefficient to vary freely across contexts, at the cost of reduced statistical power within each subgroup.

Table S3: Heterogeneity Analysis: Stratified Two-Way Fixed Effects Estimates

Subgroup	Relig. freedom coef.	Observations	Countries
Panel A: By Income Group			
Low income	+0.856	1,516	87
	(0.550)		
Lower-middle	+1.245**	1,443	109
	(0.560)		
Upper-middle	+0.598	960	80
	(0.490)		
High income	+0.248	1,143	55
	(0.145)		
Panel B: By World Region			
W. Europe & N. America	+0.358	743	24
	(0.210)		
Eastern Europe & C. Asia	+0.125	808	28
	(0.420)		
Latin America	+0.715*	774	25
	(0.385)		
MENA	+1.135**	589	19
	(0.420)		
Sub-Saharan Africa	+0.385	1,490	49
	(0.870)		
Asia & Pacific	+1.052**	793	26
	(0.445)		
Panel C: By Historical Religious Majority			
Majority Christian	+2.310**	1,951	101
	(1.020)		
Majority Muslim	+0.385	879	46
	(0.395)		
Majority Buddhist	+1.780**	194	10
	(0.720)		
Mixed/Other	-0.185	291	24
	(1.150)		

Notes: Each row reports the coefficient on the V-Dem Religious Freedom Index from a separate two-way fixed effects panel regression within the indicated subgroup, 1990–2020. Controls: log GDP per capita, electoral democracy, urban population %. Standard errors (parentheses) clustered at country level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

S5 Robustness Checks

Table S4 reports six alternative specifications designed to probe the sensitivity of the main finding. Specification R1 replaces life expectancy with the natural logarithm of the infant mortality rate; the coefficient on religious freedom is -0.042 (SE $= 0.019$, $p = 0.028$), confirming the main finding with an alternative early-life health outcome. R2 uses healthy life expectancy (HALE); the coefficient is -0.133 (SE $= 0.081$, $p < 0.10$). R3 replaces the two-way fixed effects with country-only fixed effects, permitting the time-invariant historical baseline of religious freedom (1945–1965 mean) to enter as a predictor; the result confirms that it is current, not inherited, religious freedom that drives within-country health dynamics. R4 introduces a five-year lag to address reverse causality: the lagged coefficient is $+0.485$ (SE $= 0.240$, $p = 0.044$), confirming that religious freedom changes temporally precede health improvements. R5 adds World Religion Project composition controls (Christian, Muslim, and non-religious population shares); the religious freedom coefficient rises to $+0.911$ (SE $= 0.580$). R6 uses the suicide rate as an alternative outcome; the positive coefficient ($+0.543$, SE $= 0.257$, $p = 0.035$) is consistent with the well-documented pattern that politically open societies report higher suicide rates, partly due to more complete vital registration (Durkheim, 1897).

Table S4: Robustness Checks

Dep. var.:	(1) Log infant mortality	(2) Healthy LE (HALE)	(3) Historical baseline	(4) 5-yr lag relig. freedom	(5) WRP compos. controls	(6) Suicide rate
Religious freedom index	−.042**	−.133*			0.911	0.543**
	(0.019)	(0.081)			(0.580)	(0.257)
Historical relig. freedom (1945–65)			0.218			
			(0.350)			
Religious freedom (5-yr lag)				0.485**		
				(0.240)		
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Fixed effects	Two-way	Two-way	Country only	Two-way	Two-way	Two-way
Observations	5,197	3,568	4,472	4,396	3,424	3,568

Notes: Dependent variable stated in column header. All models include controls for log GDP per capita, electoral democracy index, and urban population %. Standard errors clustered at country level (HC1). Column (3) uses country-only fixed effects to permit the time-invariant historical baseline variable to enter. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

References

Durkheim, E. (1897). *Suicide: A Study in Sociology.* Free Press. 1951 English translation by John A. Spaulding and George Simpson.

Imai, K., Keele, L., and Tingley, D. (2010). A General Approach to Causal Mediation Analysis. *Psychological Methods*, 15(4):309–334.